Limbayong

Deepwater oil and gas development · Block G · Offshore Sabah

Working Draft v0.2 — Parts 1 and 3, with ledger, source register and build status

Supersedes v0.1. Issued in response to an independent audit of the previous revision.

Document ID	LIM-SFDP-v0.2
Revision	0.2 — rebuilt on the as-built vessel basis
Date	19 October 2026
Prepared by	HERMES (arifOS). Not attributed to any person.
Status	Working draft — two Parts of twelve. Not a review pack.
Scope	Part 1, Part 3, Appendices B, E, F, G
Classification	Independent reconstruction from public sources

NOT A PETRONAS DOCUMENT

This is not a PETRONAS or PETRONAS Carigali document, and was not produced with, reviewed by, or endorsed by either company, MISC, or any contractor. It contains no internal data. No PETRONAS logo, branding or document numbering is used or implied.

Decision frame

For a reader with two minutes. Every figure is sourced in Appendix B and E.

> **On the vessel it actually bought, Limbayong produces 25,000 barrels a day — not 40,000.**

Every planning figure in this programme that assumed 40,000 bopd, 75,000 barrels a day of water injection, and 600,000 barrels of storage was reading a 2020 FPSO tender that was never awarded. The vessel that was bought — the **Excalibur**, IMO 9417103, ex-**Sevan Hummingbird**, built 2007 — is published by its owner at 25,000 bopd, 30,000 barrels of total liquids, 20,000 barrels a day of water injection, and 270,000 barrels of working storage.

What the resource gives

139.5 MMbbl at P50, from a declared bracket of 347–398 MMbbl oil in place and 35–40 per cent recovery. DERIVED

What the vessel allows

25,000 bopd. At that rate the resource depletes in 15.3 years, P90–P10 range 14.5–16.1. VERIFIED

What the gap is

15,000 bopd of capacity that either exists after MMHE conversion, or does not. No public source says which. OPEN

Three questions a chairman can put tomorrow

1. **What is the post-conversion topsides specification, in writing?** Not the 2020 tender — the as-built vessel. Every production, injection and storage number in this document changes with the answer.
2. **Is the development phased around a 25,000 bopd hull, or does it need a second hub?** At 25,000 bopd this is a 15-year asset. At 40,000 it would be a 9.6-year asset that cannot be held. Those are different projects with different economics.
3. **What is the gas route?** Limbayong is an oil accumulation and a non-associated gas field — Limbayong-1 found gas in 2002, Limbayong-2 found oil in 2013. The vessel's gas handling capacity is not published at all. VERIFIED

What this pack does not answer

No net present value, no internal rate of return, no capital cost, no schedule, no economic limit. Those need a fiscal engine built on Malaysian PSC terms and a term sheet. Neither is public, and three ad-hoc fiscal models in this programme produced three different answers from the same inputs. A fourth would repeat the defect rather than resolve it.

What this pack does is fix the **physical** basis, so that the economics — when they exist — are computed on the right field.

Document control

Revision block

Rev	Date	Change
0.1	19 Oct 2026	First issue. Parts 1 and 3. Nine pages.
0.2	19 Oct 2026	Rebuilt on the as-built vessel basis after independent audit. Vessel identity resolved; facility basis split into four spec sets; corner-product cases replaced by a correlated Monte Carlo; VRR corrected; gas ratio relabelled; water depth reclassified; figure QA automated.

What changed, and why

Item	Change in v0.2
Vessel identity	Excalibur resolved to IMO 9417103 = Sevan Hummingbird = Hummingbird Spirit. v0.1 treated it as an unnamed hull.
Oil capacity	30,000 bopd in v0.1 → **25,000 bopd** as-built, with 30,000 bpd as total liquids.
Water injection	75,000 bwpd in v0.1 → **20,000 bwpd**. The 75,000 was a tender spec.
Storage	300,000 bbl “versus spec” → **300,000 design / 270,000 working**. Not a discrepancy.
Cases	Low/Mid/High corner products → correlated Monte Carlo reporting P90/P50/P10.
Gas	5,640 scf/bbl presented as a GOR → relabelled a **ratio of totals**, withdrawn from gas-train sizing.
Water depth	900–1,400 m, tagged OPEN → **900–1,200 m**, two sources; upper bound withdrawn.
Cost anchors	Absent → subsea award bands added as L-38 and L-39.
Status	“Review pack” → **working draft, two Parts of twelve**.

Reading guide — five tags

Tag	Test it must pass
VERIFIED	Two or more independent sources, or one primary document (a filing, a statute, a contract). Independence is counted per row in Appendix B.
REPORTED	One source only, URL given. Carries the single-source flag; a critical-path number in this class is a risk, not a fact.
DERIVED	Computed here from other ledger rows. The formula is in the row, so the arithmetic can be repeated without trusting the author.
ASSUMED	A judgement, not a measurement. Owner and reason stated. Can be challenged; cannot be verified.
OPEN	The input does not exist in the public record. Marked, never guessed. Appears as a red banner where it changes a conclusion.

Two rules this document is built on. First, no number may exist only in prose — every figure traces to a row in Appendix B. Second, no conflict is resolved silently: where two sources disagree, both are printed and the disagreement is given an ID.

Data classification

All information derives from public sources: one Bursa Malaysia filing, trade press, an operator-side project description, a non-commercial resource tracker, and public-domain geospatial data. No PETRONAS, PETRONAS Carigali, MISC or contractor confidential material has been used, requested or received. A reader with access to operator data should expect this document to differ from it, and should prefer the operator’s data.